Neuronal plasticity models and their function

Spyridon Chavlis, PhD

Athanasia Papoutsi, PhD

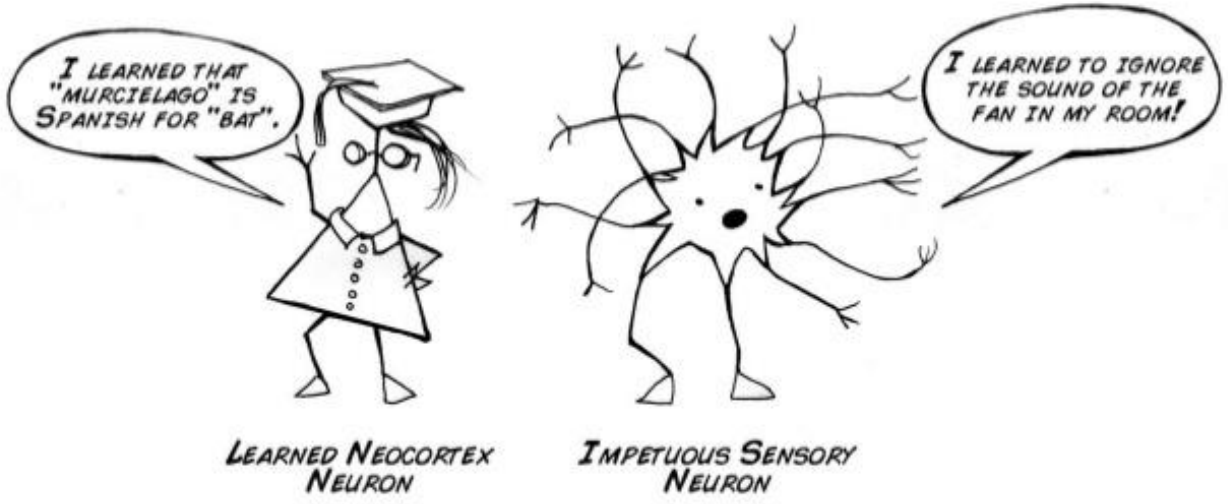
IMBB-FORTH

Based on:
Neuronal Dynamics: From single neurons to networks and models of cognition
Wulfram Gerstner, Werner M. Kistler, Richard Naud and Liam Paninski
https://neuronaldynamics.epfl.ch/

Brains adapt

Brains produce learned or adaptive behaviors that allow animals to thrive in complex environments.

Accurate perception of the sensory elements of an environment, an ability to **move appropriately** within the environment and a capacity to make **suitable decisions**.



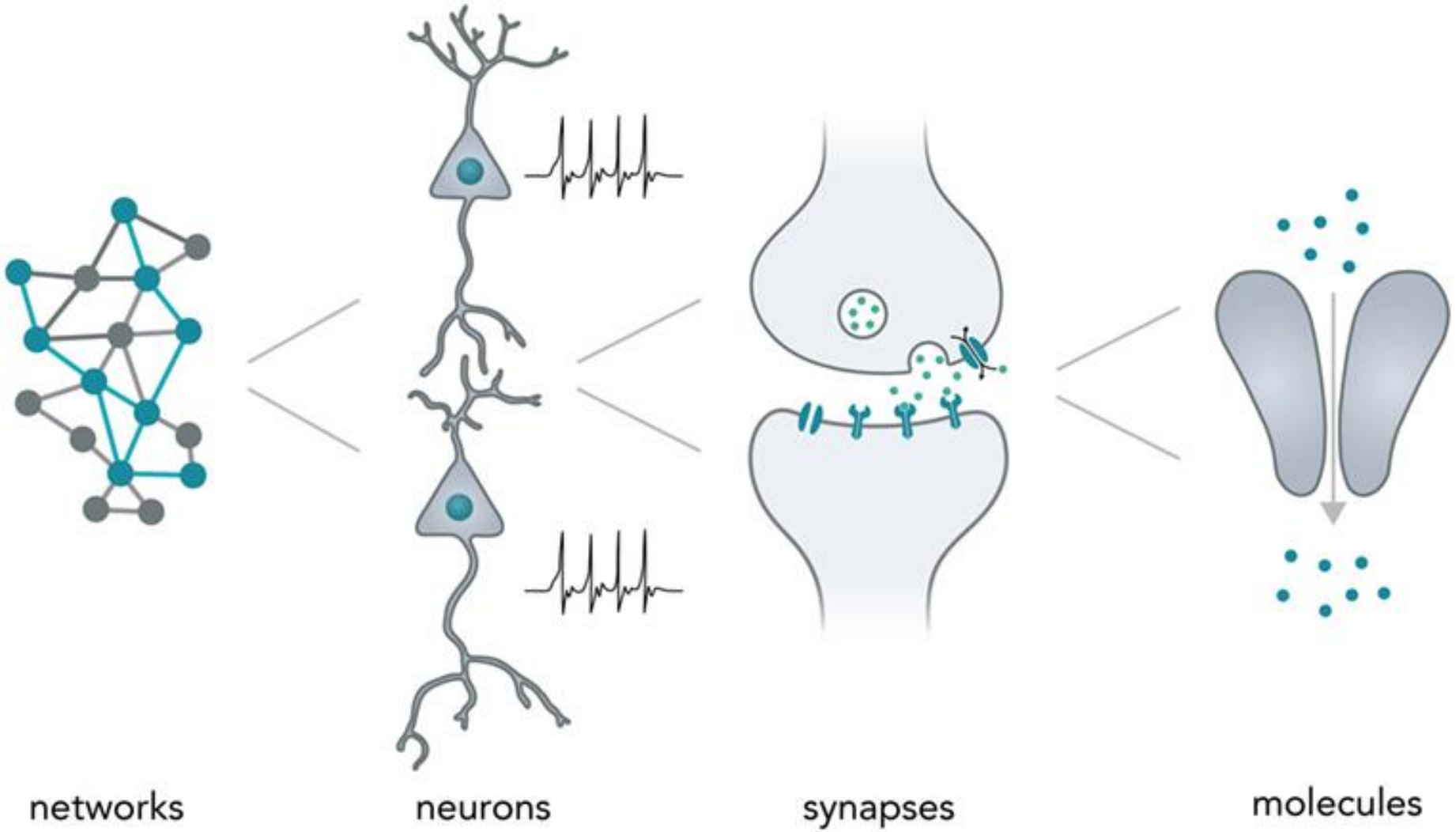
Brains adapt

The process of acquiring a particular adaptive behavior involves modifications to many brain regions, with the actual site of the adjustments being found within the individual neurons that make up these regions.

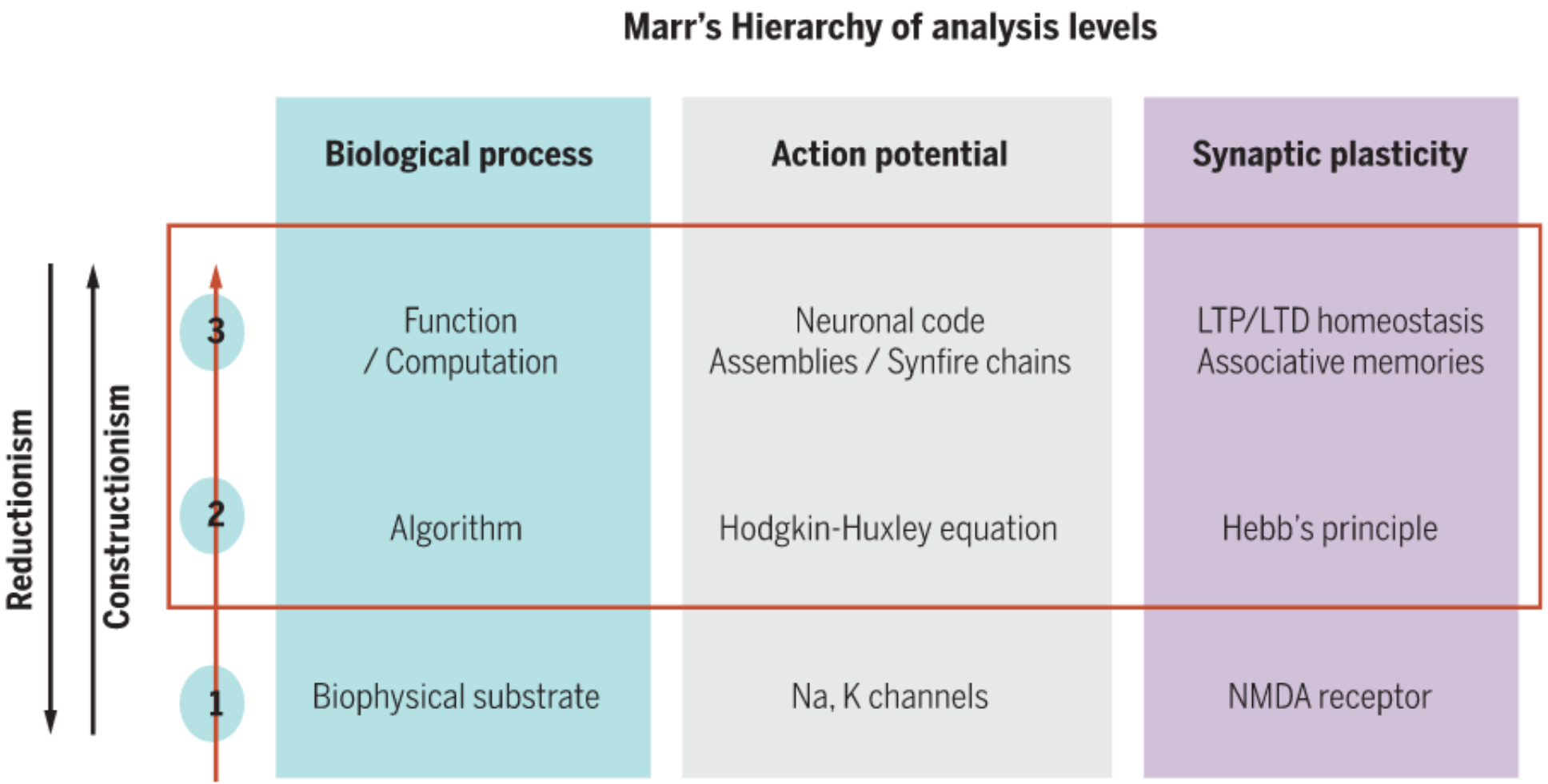
Adaptations: changes in the responsiveness of the neurons to particular input patterns =>

Adaptive changes **alter the input-output transformations of neurons** within a given region so that the population can best perform the specific computations that enable successful behaviors.

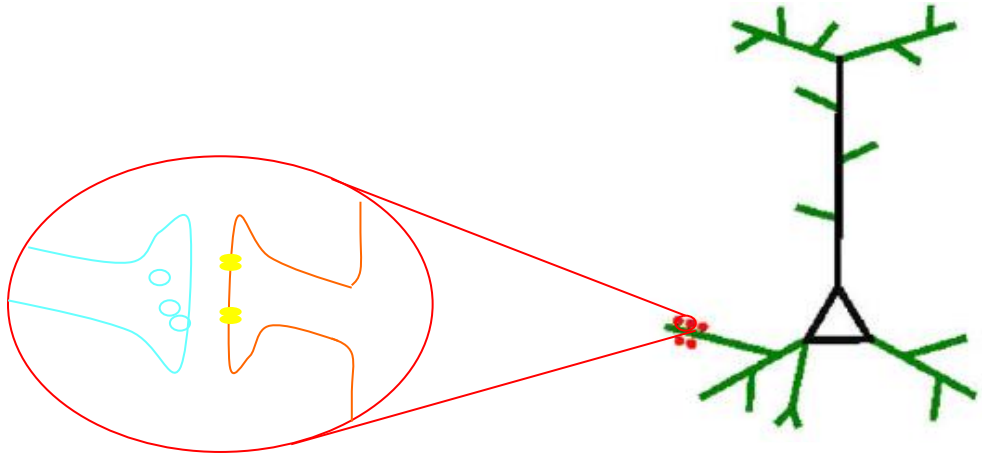
Synaptic plasticity and learning

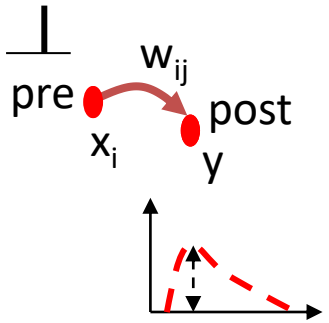


Marr's 3 levels of analysis



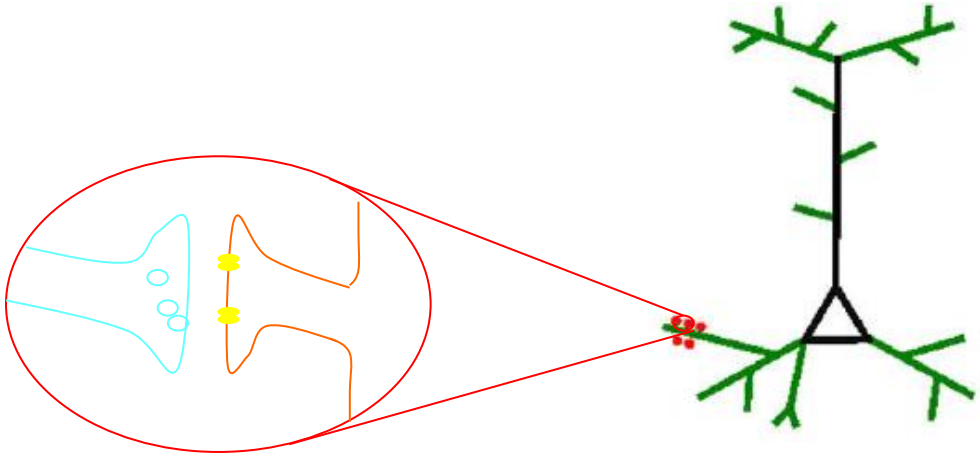
Synaptic weights

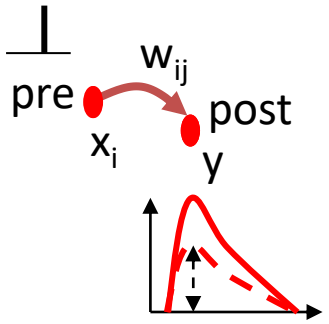




Each synapse is characterized by a single constant parameter $\boldsymbol{w_{ij}}$, called the synaptic weight.

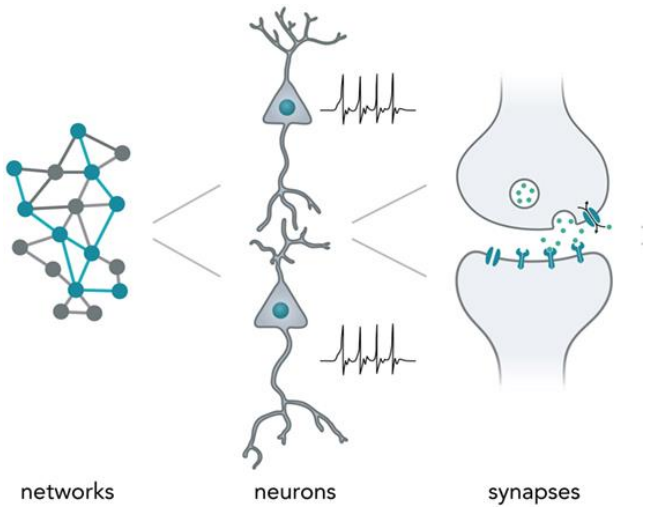
Synaptic weights

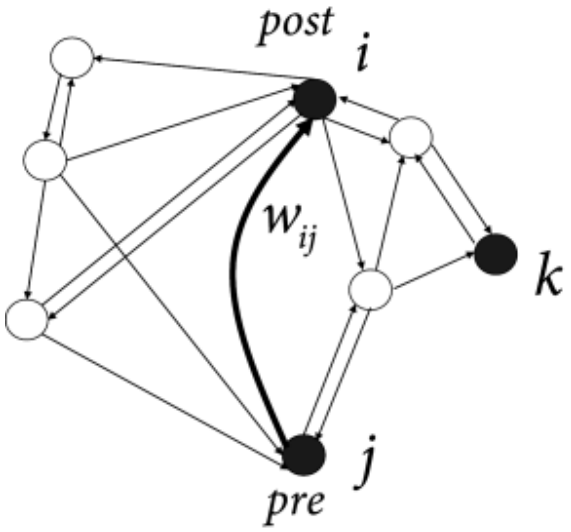




Synaptic weights

The weight w_{ij} of a connection from neuron j to i is considered a parameter that can be adjusted so as to optimize the performance of a network for a given task. The process of parameter adaptation is called **learning** and the procedure for adjusting the weights is referred to as a **learning rule**.





Types of Learning

- Unsupervised, supervised, reward-based

- Rate-based, spike-based

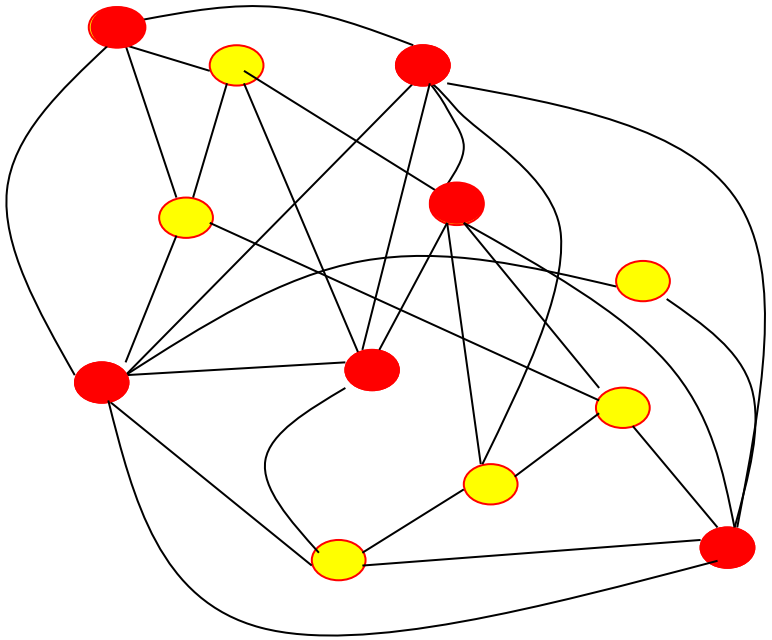
Hebb rule (1949)

- If cell A “repeatedly or persistently takes part” in firing cell B, then the strength of their connection should increase.
- Neurons that ‘fire together, wire together’
- Correlation-based learning

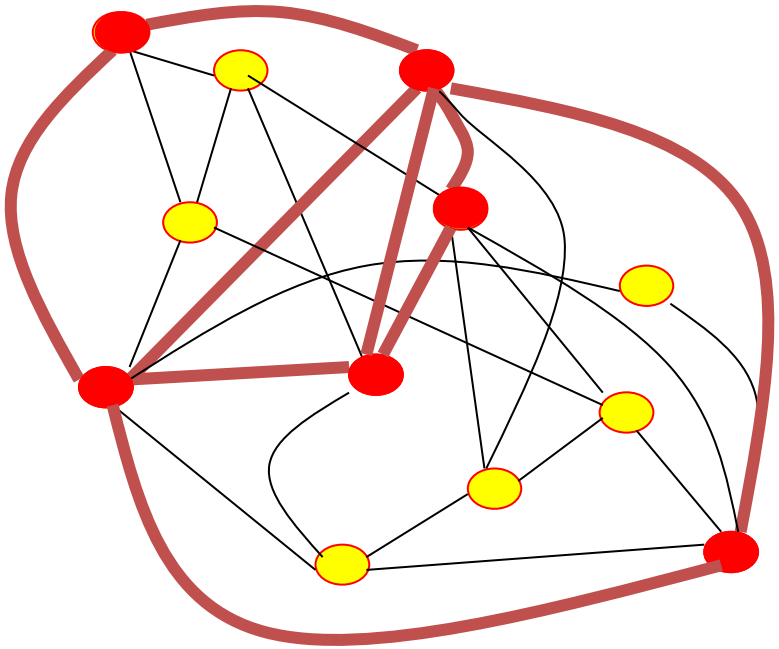
Donald Hebb

No notion of ‘good’ or ‘bad’ changes of a synapse, Unsupervised learning!

Hebbian learning

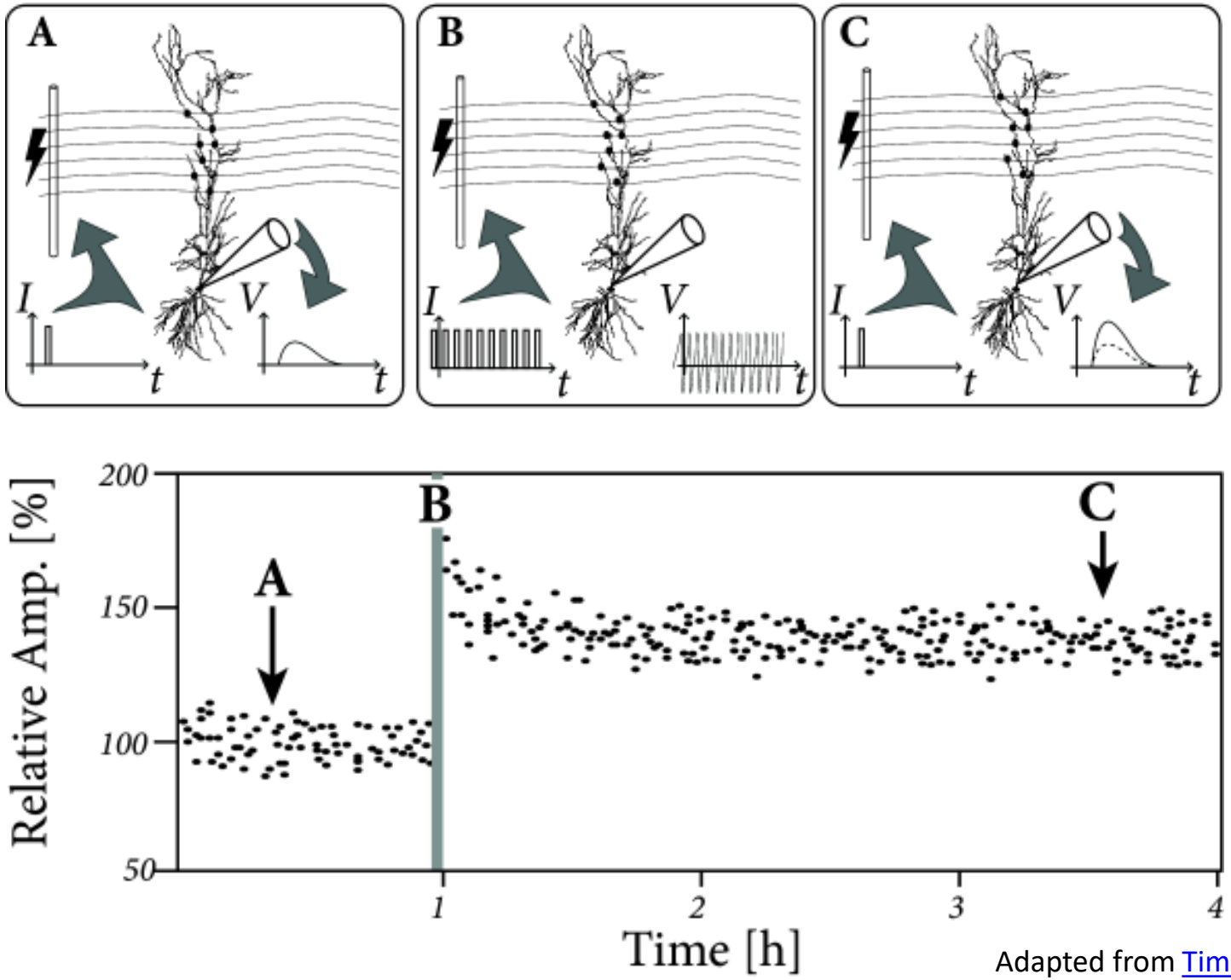


Hebbian learning



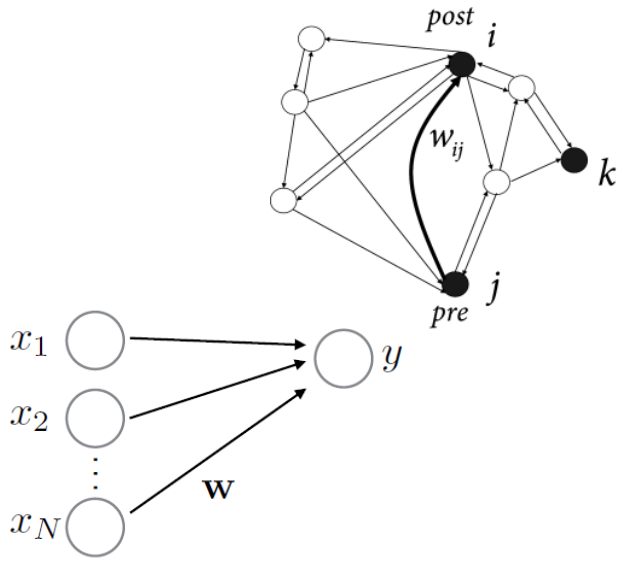
item memorized

Long-term **potentiation**



Adapted from [Timothy Bliss and Terje Lømo, 1973]

Models of Hebbian learning



Unsupervised, rate-based

Locality: change of the synaptic efficacy can only depend on local variables

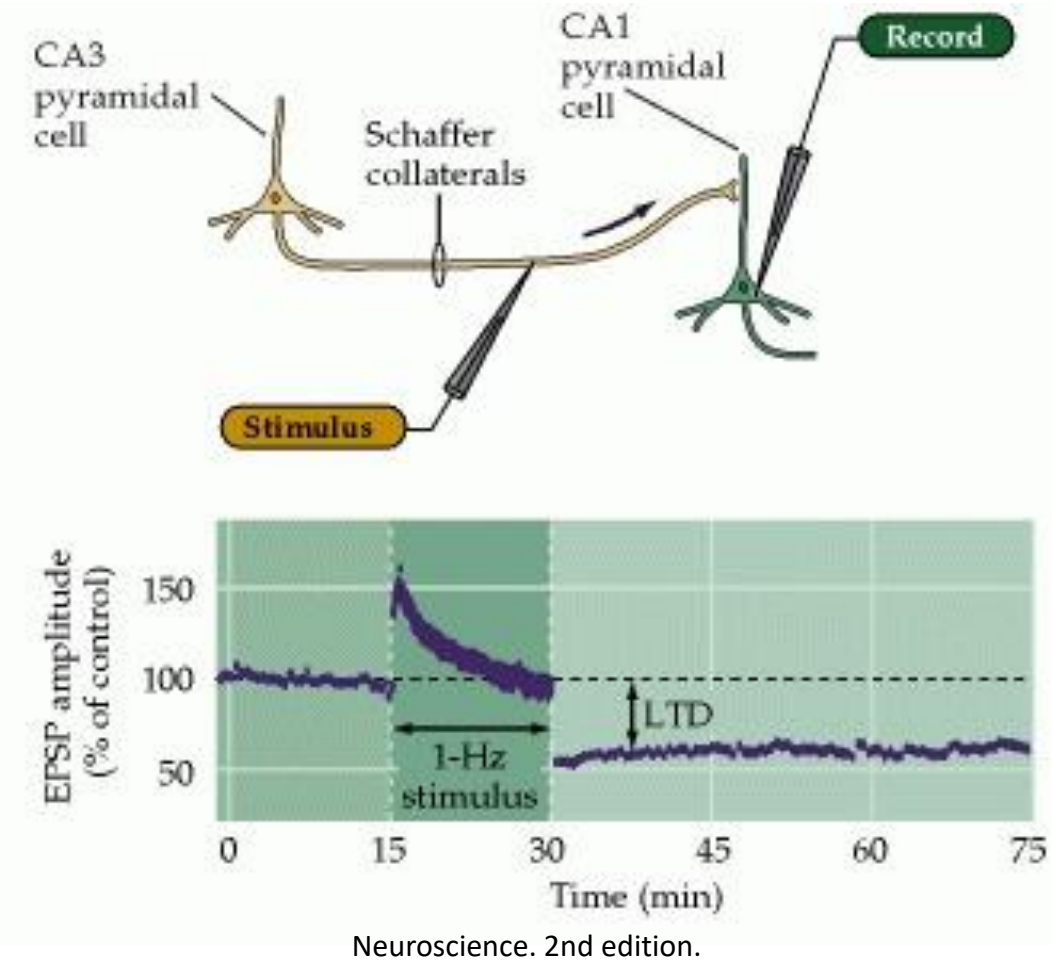
$$\frac{dw_{ij}}{dt} = F(w_{ij}, x_i, y)$$

Joint activity: pre- and postsynaptic neurons have to be active simultaneously for a synaptic weight change

Simplest Hebbian rule: $\frac{dw_{ij}}{dt} = \eta x_i y$, η depends on the current value of the weight, $\eta(w_{ij})$

Long-term **depression**

LTD occurs when the Schaffer collaterals are stimulated at a low rate—about 1 Hz—for long periods (10–15 minutes)

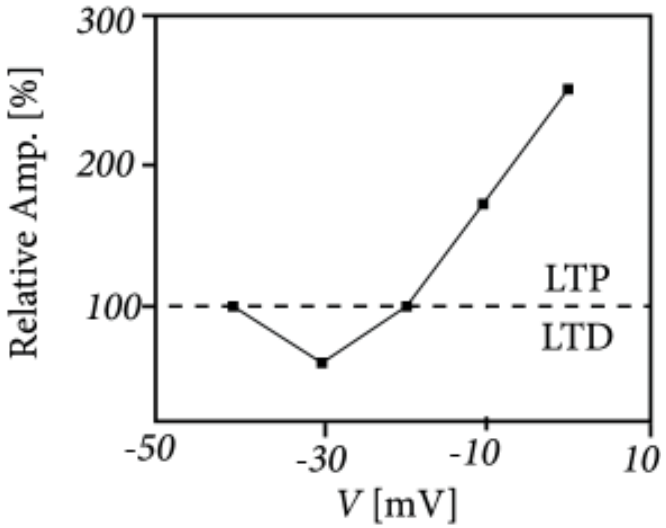


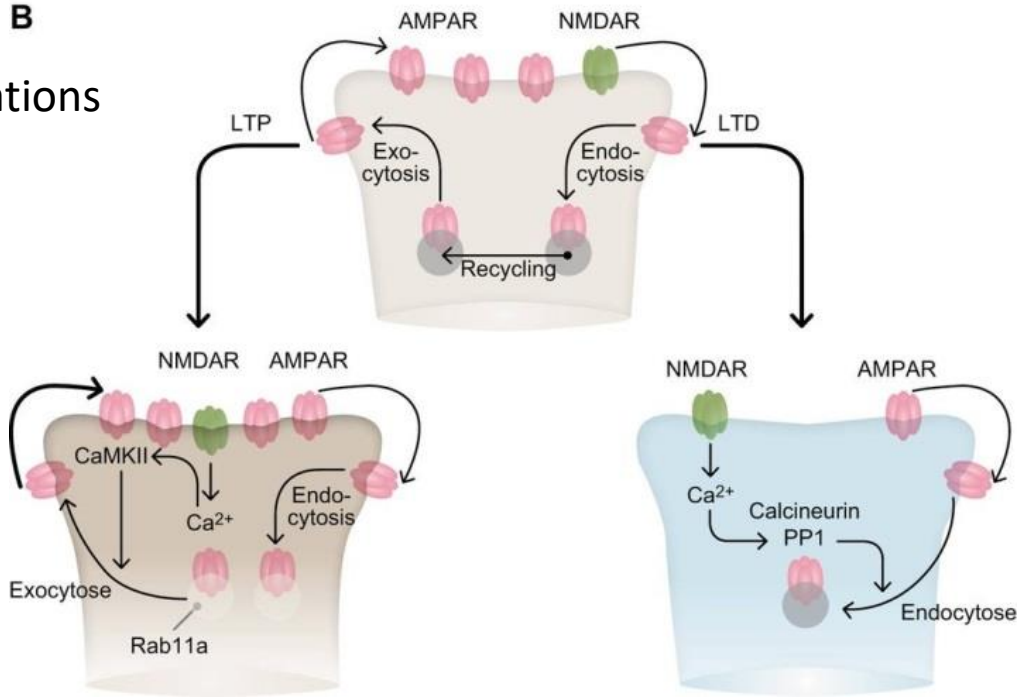
Neuroscience. 2nd edition.

Long-term potentiation / depression

Both LTP and LTD require activation of NMDA-type glutamate receptors and the resulting entry of Ca^{2+} into the postsynaptic cell. The major determinant of whether LTP or LTD arises appears to be the amount of Ca^{2+} in the postsynaptic cell: small rises in Ca^{2+} lead to depression, whereas large increases trigger potentiation.

Voltage dependence of synaptic modifications

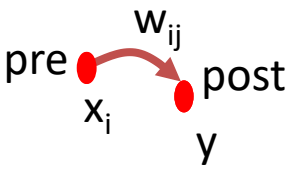




Adapted from Korte and Schmitz , 2016

Models of Hebbian learning

$$\frac{dw_{ij}}{dt} = \eta x_i y$$

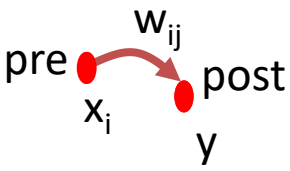


pre	on	off	on	off
post	on	on	off	off
	+	**0**	**0**	**0**

Models of Hebbian learning

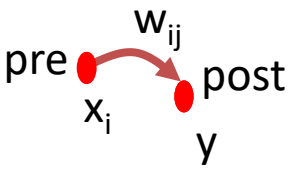
$$\frac{dw_{ij}}{dt} = \eta x_i y$$

$$\frac{dw_{ij}}{dt} = \eta x_i y\text{-c}$$



pre	on	off	on	off
post	on	on	off	off
	+	0	0	0
	+	-	-	-

Models of Hebbian learning



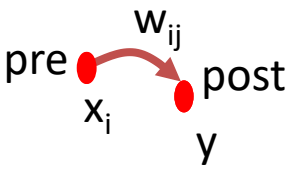
$$\frac{dw_{ij}}{dt} = \eta x_i y$$

$$\frac{dw_{ij}}{dt} = \eta x_i y\text{-c}$$

$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

pre	on	off	on	off
post	on	on	off	off
	+	0	0	0
	+	-	-	-
	+	0	-	0

Models of Hebbian learning



$$\frac{dw_{ij}}{dt} = \eta x_i y$$

$$\frac{dw_{ij}}{dt} = \eta x_i y\text{-c}$$

$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

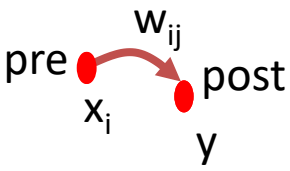
$$\frac{dw_{ij}}{dt} = \eta (x_i - \theta)(y - \theta)$$

pre	on	off	on	off
post	on	on	off	off
	+	0	0	0
	+	-	-	-
	+	0	-	0
	+	-	-	+

$$\frac{dw_{ij}}{dt} = \gamma(x_i - < x_i >)(y - < y_i >)$$

The covariance rule (*Sejnowski , 1999)*

Models of Hebbian learning



$$\frac{dw_{ij}}{dt} = \eta x_i y$$

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$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

$$\frac{dw_{ij}}{dt} = \eta (x_i - \theta)(y - \theta)$$

pre	on	off	on	off
post	on	on	off	off
	+	0	0	0
	+	-	-	-
	+	0	-	0
	+	-	-	+

$$\frac{dw_{ij}}{dt} = \gamma(x_i - < x_i >)(y - < y_i >)$$

The covariance rule (*Sejnowski , 1999)*

Models of Hebbian learning

$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

Δw_{ij} over y, assuming $x_i > 0$?

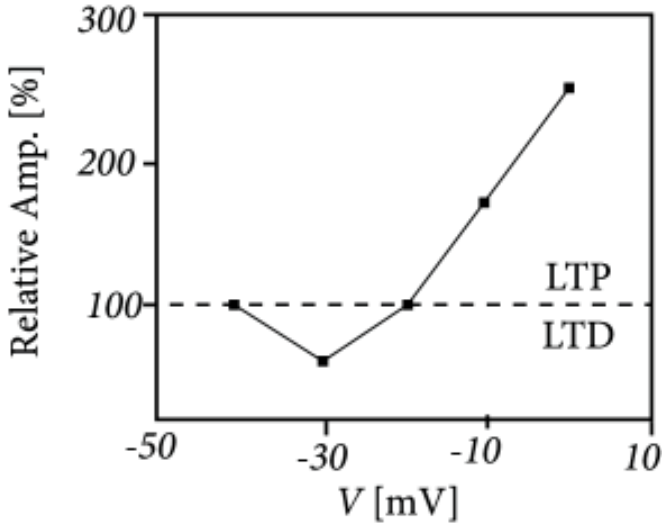
Models of Hebbian learning

$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

Δw_{ij} over y, assuming $x_i > 0$?

For $y=\theta$, $\Delta w_{ij} = 0$

For $y=0$, $\Delta w_{ij} = -\eta x_i \theta$



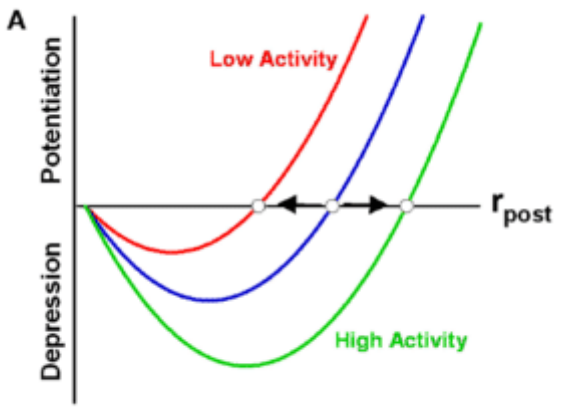
Models of Hebbian learning

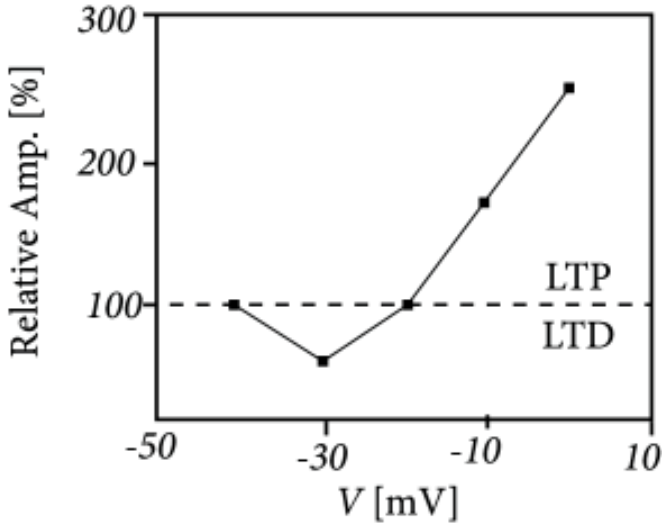
$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta)$$

Δw_{ij} over y, assuming $x_i > 0$?

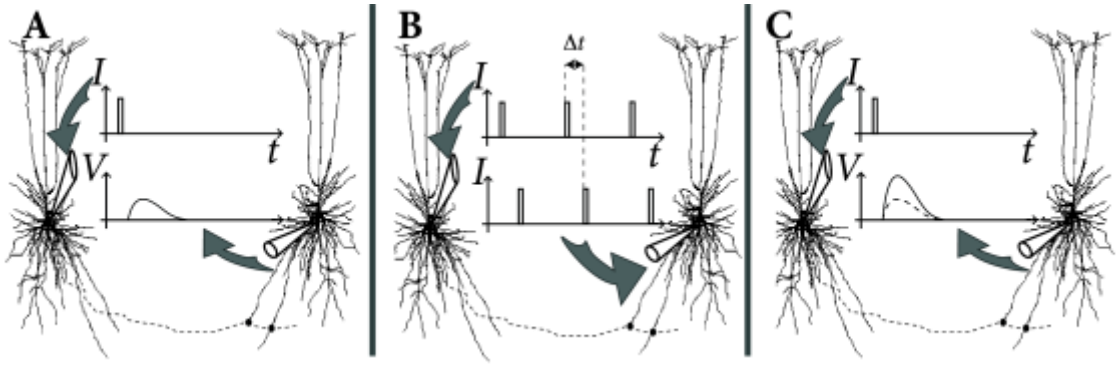
Bienenstock, Cooper, Munro (BCM rule):

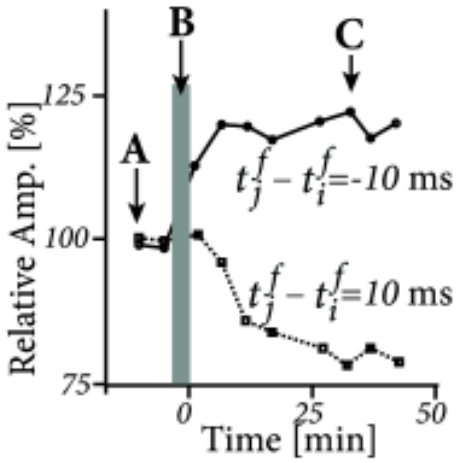
$$\frac{dw_{ij}}{dt} = \eta x_i (y - \theta) y$$

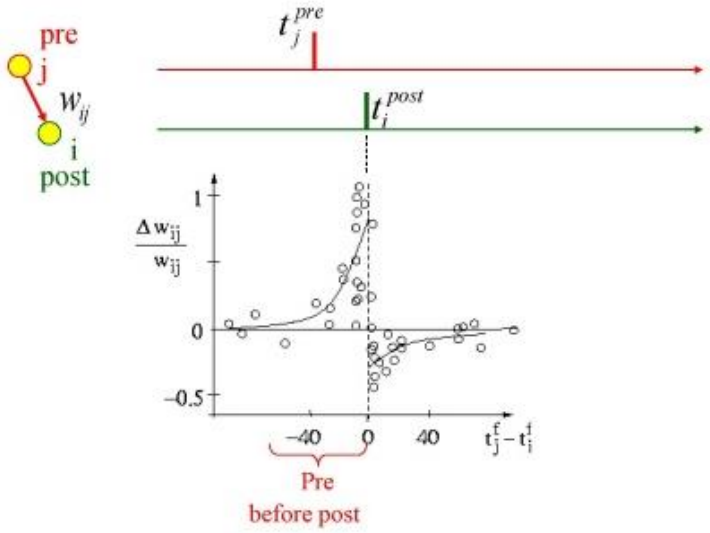




Spike-timing-dependent plasticity (STDP)

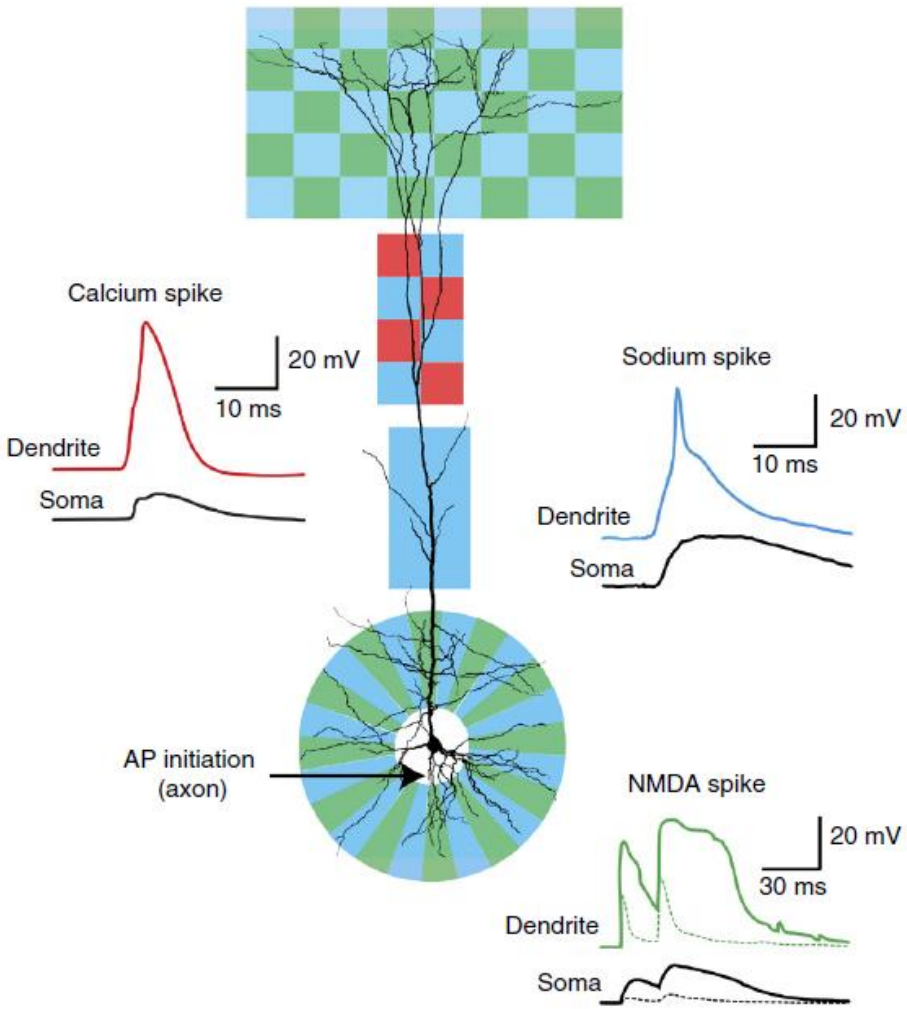




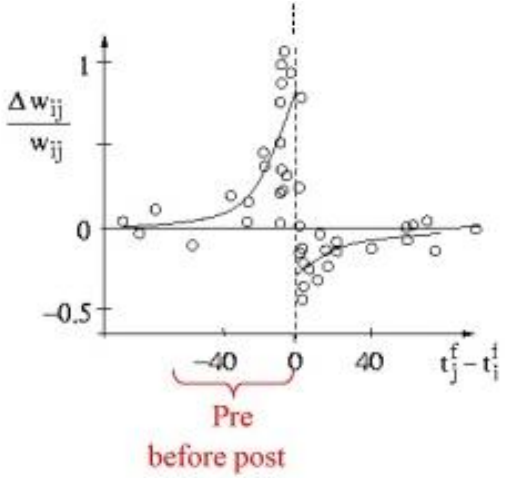


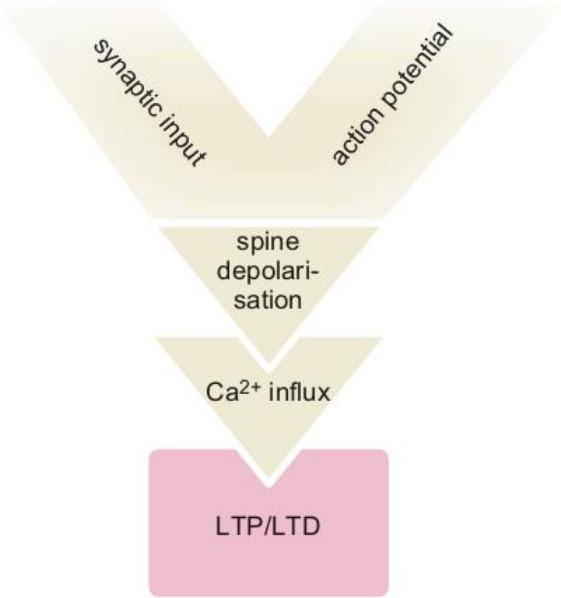
In agreement with Hebb's postulate because presynaptic neurons that are active slightly before the postsynaptic neuron are those which 'take part in firing it'

Spike-timing-dependent plasticity (STDP)



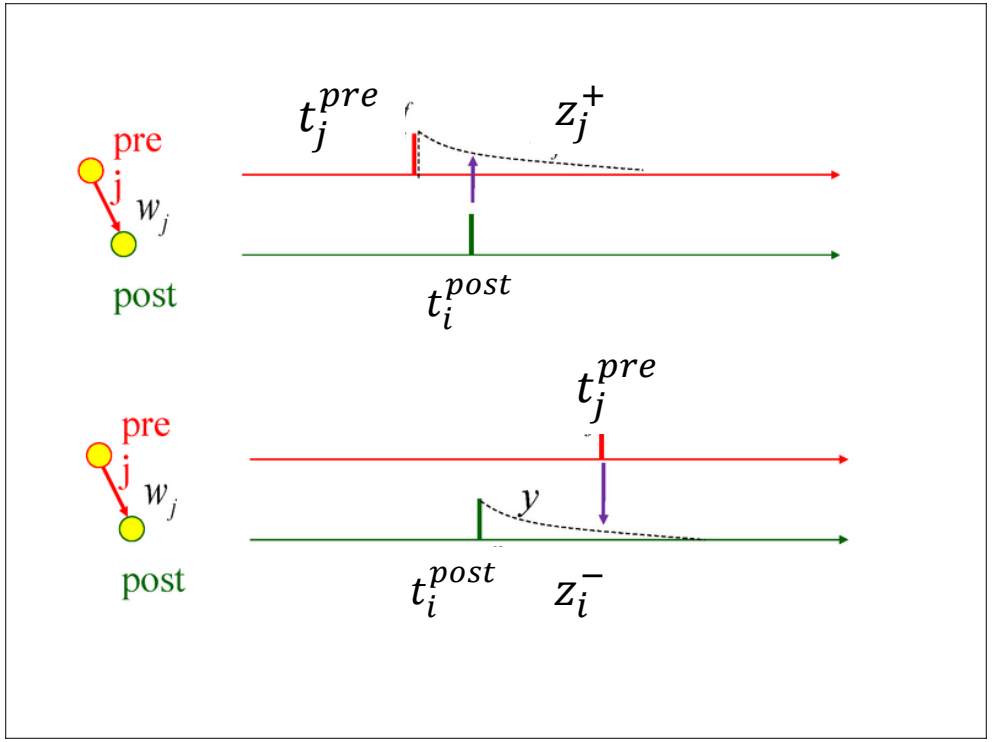
Stuart, Spruston, Nature Reviews, 2015

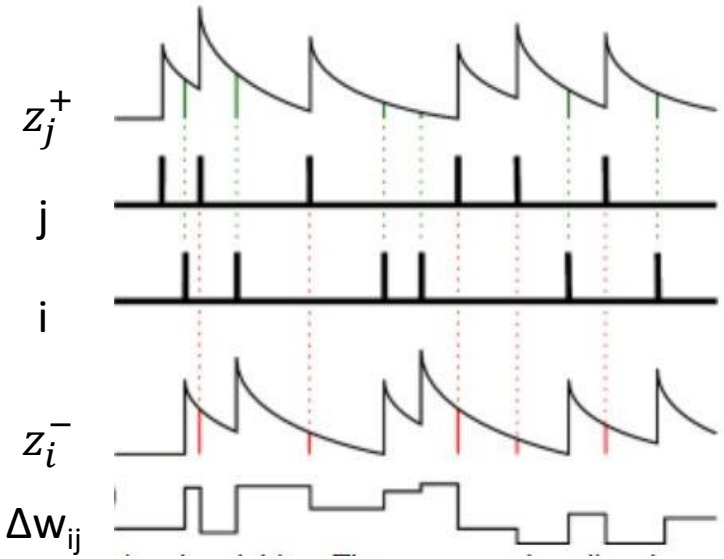




Korte , Schmitz , 2016

Spike-based models





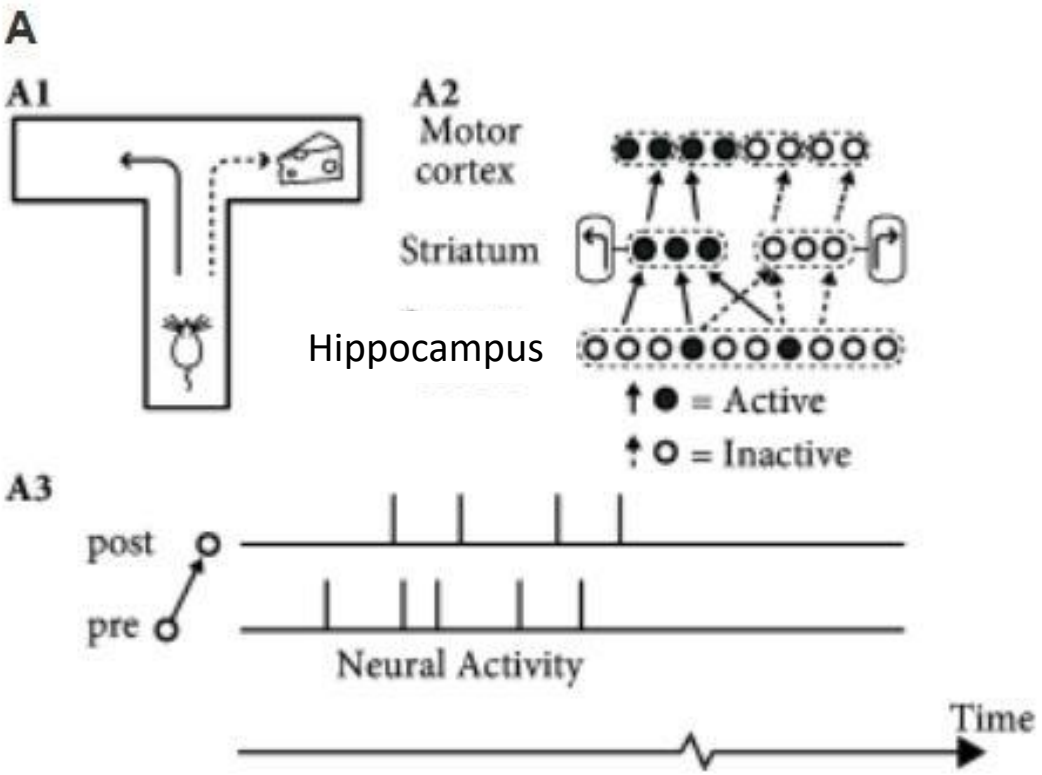
$$\frac{dW_{ij}}{dt} = A_+(w_{ij})z_j^+\delta(t - t_i^{post}) - A_-(w_{ij})z_i^-\delta\left(t - t_j^{pre}\right)$$

Problematics of Hebbian learning

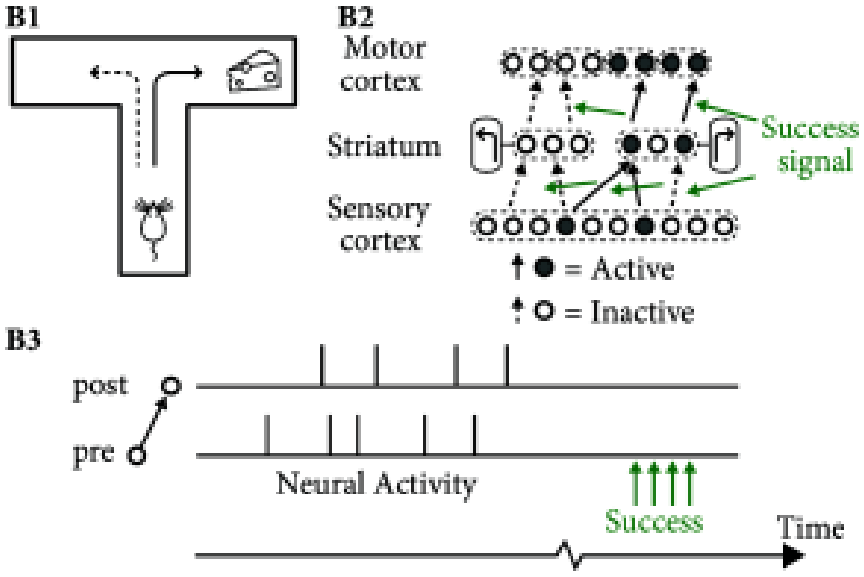
- It is difficult for synaptic plasticity mechanisms operating on physiological timescales (tens of milliseconds) to handle the time course of behavior (at least seconds).

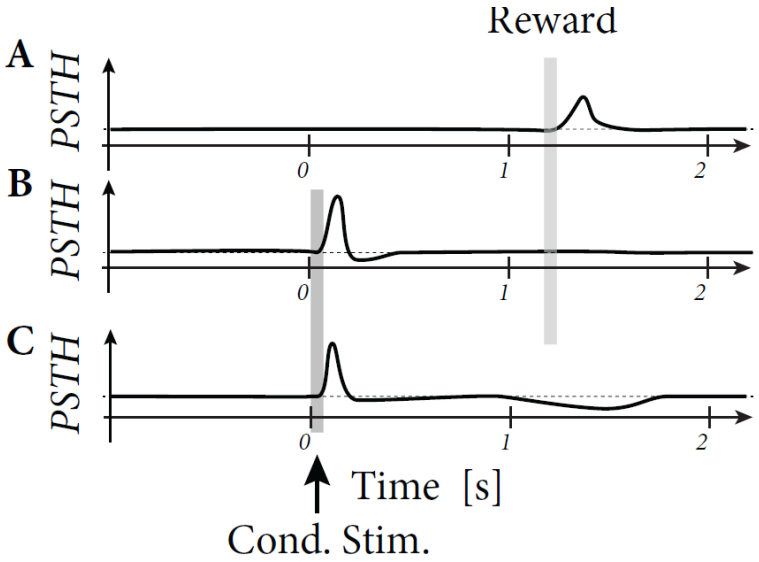
- Hebbian learning lacks a target for neuronal activity to approach, thus limiting these self-organizing mechanisms to the formation of fundamental types of representations that are not always sufficient for achieving a specific learning objective.

Reward-based learning



Reward-based learning





Dopamine encodes success=
reward – expected reward
Schultz et al., 1997

Two additional aspects:

1. Success of an action.
2. The success often comes with a delay of a few seconds after an action has been taken.

Three-factor learning rule:

$$\frac{dw_{ij}}{dt} = F(pre, post, Modulatory\ signal)$$

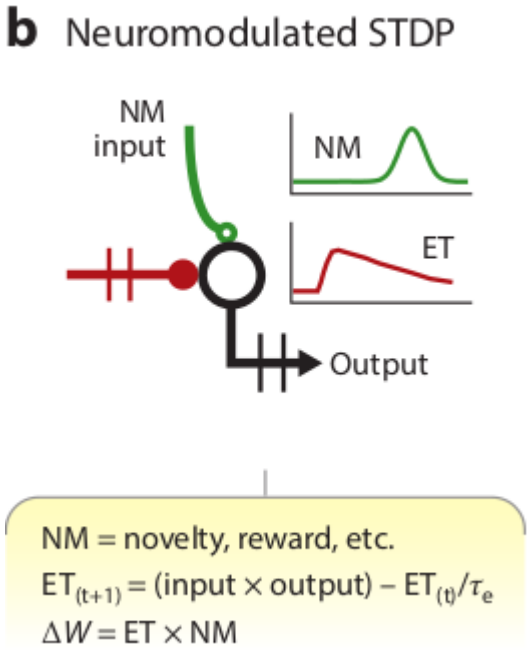
local global

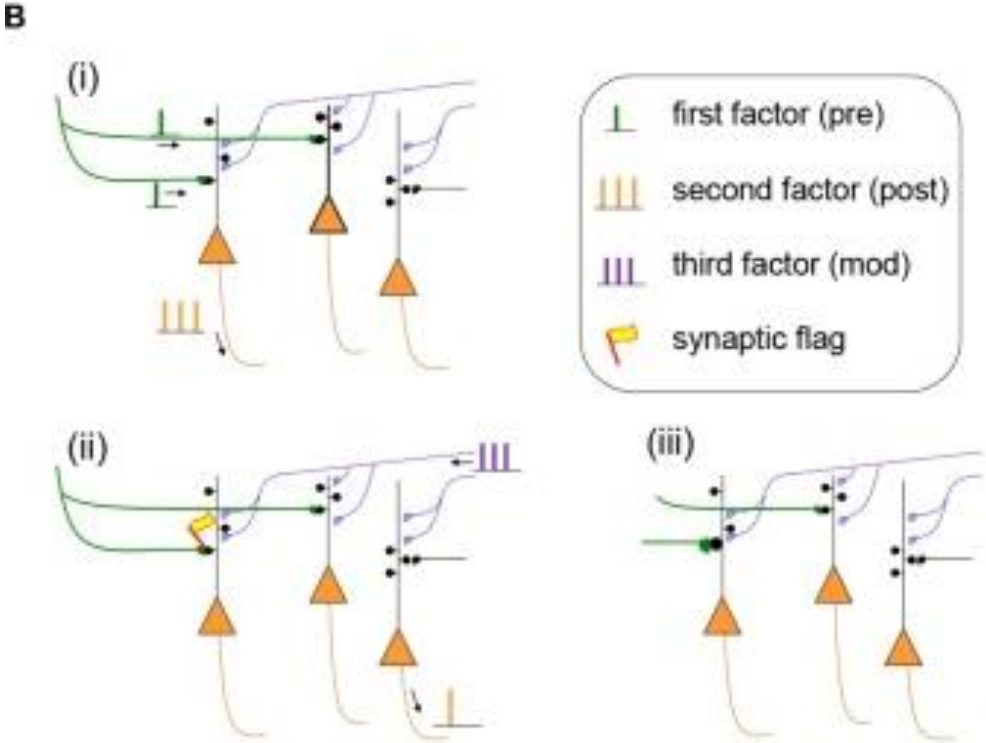
Reward-based learning

Three-factor learning rule: $$\frac{dw_{ij}}{dt} = F(pre, post, Modulatory\ signal)$$

Synapse characteristics:

1. Synaptic strength w_{ij}
2. Synapse-internal variable e_{ij} : represents a metastable transient state of interacting molecules in the spine which serves as a 'synaptic flag' or 'eligibility trace' indicating that the synapse is ready for an increase or decrease of its weight



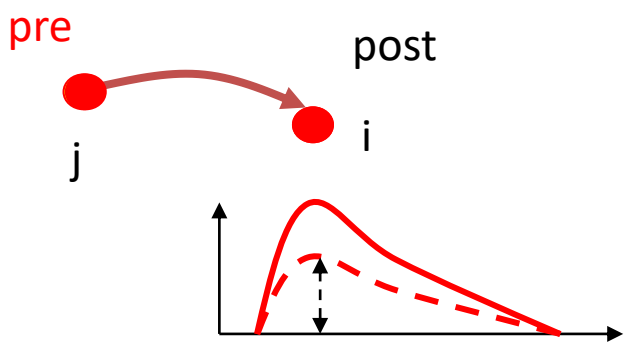


Gerstner et al. 2018

Unsupervised Vs Reinforcement

LTP/LTD/Hebb

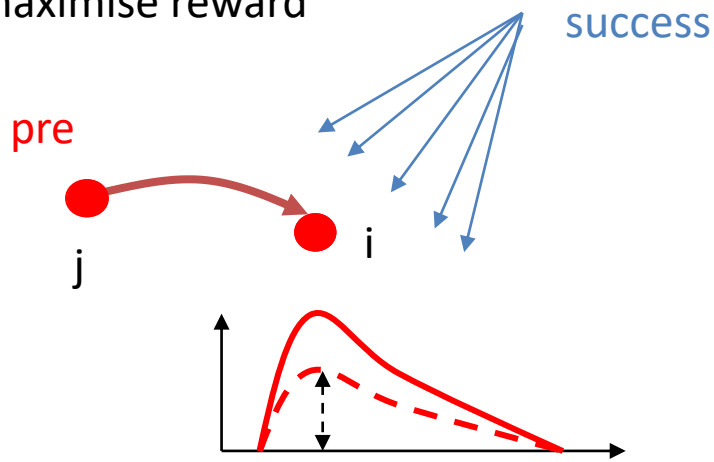
Theoretical concept
- passive changes
- exploit statistical correlations



Functionality
-useful for development
(Feature extraction and associations)

Reinforcement Learning

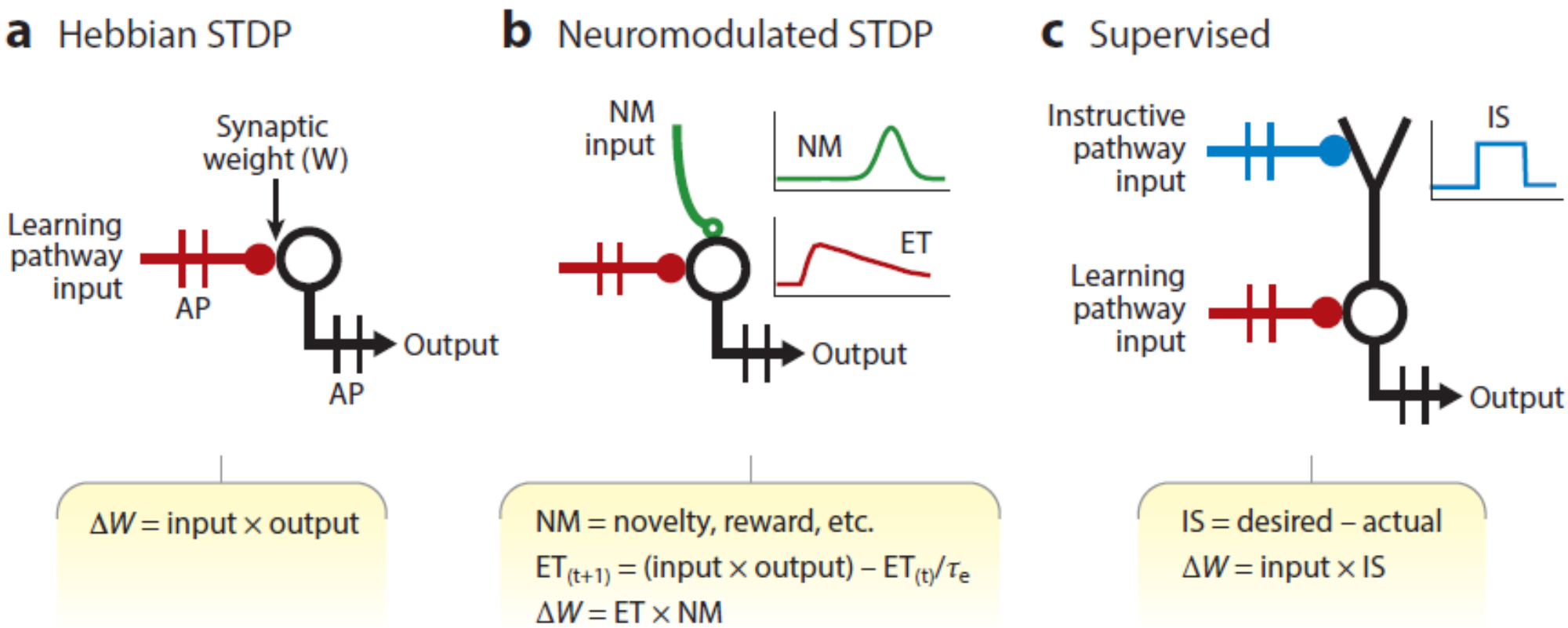
Theoretical concept
- conditioned changes
- maximise reward



Functionality
- useful for learning a new behavior

Adapted from: [Computational Neuroscience: Neuronal Dynamics of Cognition]

All together and some more…



Adapted from Magee and Grienberger, 2020